AgriTrace: Decentralized Farm-to-Market Traceability

A Blockchain-Enabled Transparent Price Discovery Architecture

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1. ABSTRACT

Traditional agricultural supply chains in emerging markets suffer from opaque pricing intermediaries, inefficient cold-chain logging, and fraudulent provenance assertions. AgriTrace addresses these challenges by combining Ethereum-compatible smart contracts, cryptographically anchored SHA-256 metadata records, and high-resolution IoT temperature and humidity telemetry.

2. SMART CONTRACT ARCHITECTURE

The core protocol utilizes an immutable Solidity ledger deployed on Ethereum Layer-1 / Layer-2 networks. Role-based access control enforces operator privileges, ensuring only verified agricultural centers and farmers can register batch provenance. Double-spend prevention and duplicate lot hashing guarantee lot uniqueness.

3. TRANSPARENT PRICE DISCOVERY

A mathematical ratio evaluates farmer value capture percentage: Farmer Price / Consumer Price x 100. Empirical benchmarks demonstrate an increase in farmer value capture from 38% under traditional commission agents to over 78% under direct AgriTrace transparent smart contract settlements.